Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY AND MANAGEMENT, PUNE-09

Name : **Class : B.E.** **Roll No :**

PRACTICAL NO. 01

Linear Discriminant Analysis (LDA) Classification on the Iris Dataset

Aim

To implement Linear Discriminant Analysis on the Iris dataset, which finds projections that best separate the classes, and to use it to classify flower species and evaluate the result with accuracy, a confusion matrix and a classification report.

Code and Output

In [1]:
```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.metrics import (accuracy_score, confusion_matrix,
                             classification_report, ConfusionMatrixDisplay)
import matplotlib.pyplot as plt
```

In [2]:
```
iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.25, random_state=42,
    stratify=iris.target
)
model = LinearDiscriminantAnalysis()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Test accuracy: {:.4f}".format(accuracy_score(y_test, y_pred)))
print("\nConfusion matrix (rows=actual, columns=predicted):")
print(confusion_matrix(y_test, y_pred))
print("\nClassification report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names,
                            zero_division=0))
```

```
Test accuracy: 1.0000

Confusion matrix (rows=actual, columns=predicted):
[[12  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Classification report:
              precision    recall  f1-score   support

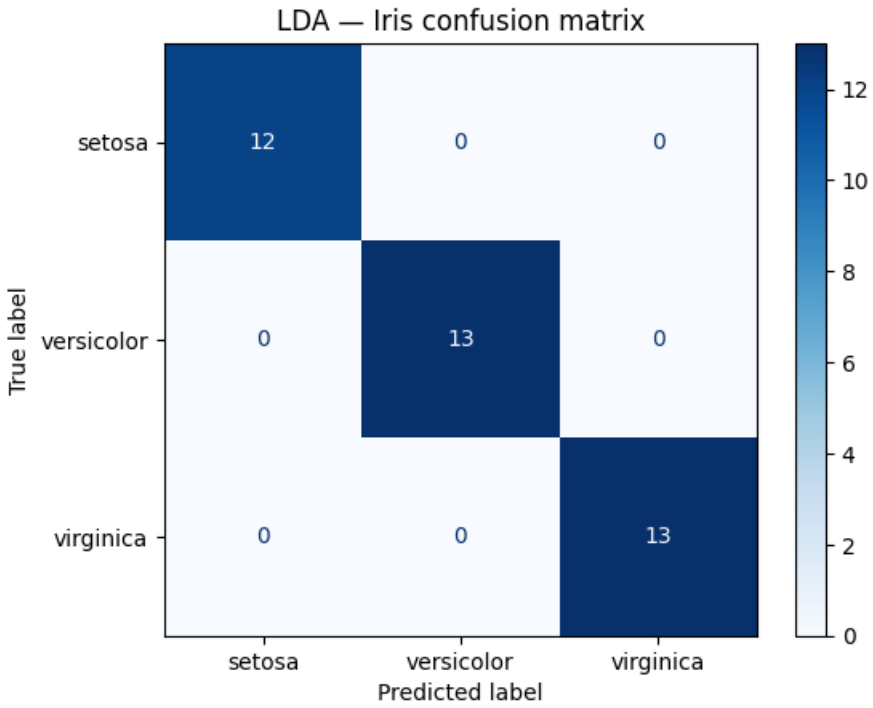
      setosa       1.00      1.00      1.00        12
  versicolor       1.00      1.00      1.00        13
   virginica       1.00      1.00      1.00        13
    accuracy                           1.00        38
```

```
   macro avg       1.00      1.00      1.00        38
weighted avg       1.00      1.00      1.00        38
```

In [3]:

```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred,
                                        display_labels=iris.target_names)
plt.title("LDA - Iris confusion matrix"); plt.show()

sample = [[5.1, 3.5, 1.4, 0.2]]
print("Sample measurements:", sample[0])
print("Predicted species:", iris.target_names[model.predict(sample)[0]])
```




```
Sample measurements: [5.1, 3.5, 1.4, 0.2]
Predicted species: setosa
```